

Rescale Pipeline Design-Space Exploration for 4-Bit Quantized MMA in Transformer FFN Layers

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I. PROJECT OVERVIEW

This project is based on **Example Project 4 (Quantization)**.

A. Problem and Motivation

LLM inference is memory-bandwidth bound: during autoregressive decoding the entire weight matrix must be streamed from HBM for every token. 4-bit weight quantization achieves a $4\times$ reduction over BF16, and all major vendors have responded—NVIDIA introduced NVFP4 with Blackwell, and the Microscaling standard (MXFP4) is gaining cross-vendor adoption.

However, these schemes are **bundled, one-size-fits-all solutions** designed for general-purpose GPUs. Each implicitly fixes the entire MMA datapath: operand format, number of rescale levels, per-level granularity, scale-factor precision, and accumulator precision. These choices are conservative to guarantee numerical safety across arbitrary workloads. For a **domain-specific accelerator targeting known workloads**, this conservatism leaves significant efficiency on the table.

We focus on **FFN/MLP layers** as the target workload because they account for approximately two-thirds of total model parameters and a comparable share of FLOPs in standard transformer architectures. During autoregressive decoding these large weight matrices dominate memory bandwidth and energy consumption, making FFN the primary efficiency bottleneck for 4-bit quantization.

Our preliminary AccelForge modeling of the NVFP4 W4A4 pipeline (Figure 1) reveals that the **rescale stages alone account for 67% of total energy**. The core FP4 multiply contributes only 33%, while quant/scale preprocessing is negligible ($<0.2\%$). This motivates exploring whether cheaper rescale configurations can preserve accuracy at substantially lower cost.

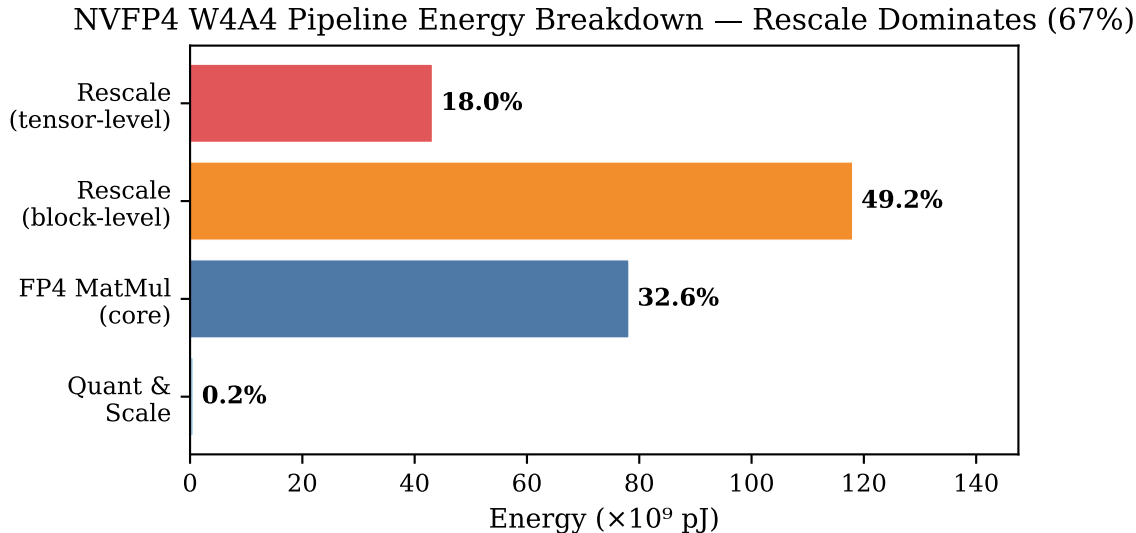


Fig. 1. NVFP4 W4A4 energy breakdown by pipeline stage. Rescale operations (block-level + tensor-level) dominate at 67% of total energy, while the core FP4 MatMul accounts for only 33%.

Recent works have made isolated improvements along individual pipeline dimensions—HiFloat4 [1] proposes a fixed three-level scaling hierarchy, AXE [2] treats accumulator precision as tunable, MixPE [3] explores dequantization placement, and “Is Finer Better?” [4] analyzes block-size/scale-format interactions. However, each produces a single new hand-designed point in the space, fixing all other dimensions. **The problem has never been formulated as a joint design-space exploration (DSE)**: no prior work decomposes these dimensions into an orthogonal search space and systematically evaluates the accuracy–hardware-cost tradeoff across configurations.

B. Key Questions

Core question: Can we formulate the design of a 4-bit MMA rescale pipeline as a structured DSE problem—defining the search space (orthogonal pipeline dimensions), the search method (systematic sweep with pruning), and the evaluation metric (joint accuracy–hardware–cost)—such that this formulation is both *valid* (contains configurations dominating all existing bundled schemes) and *necessary* (the optimal configuration is workload-specific, so no single bundled scheme can be universally optimal)?

Our contribution is not a single better configuration, but **the problem formulation itself**: the decomposition of the bundled pipeline into a tractable search space, paired with a hardware-aware evaluation framework. The finding that optimal configurations are workload-specific validates this formulation—it proves that hand-designed one-size-fits-all schemes inherently leave efficiency on the table.

C. Hypotheses

- 1) **FP32 rescale is over-provisioned** for many FFN GEMMs, particularly those with moderate accumulation chain lengths ($K \leq 4096$) and well-behaved weight distributions. Lower-precision rescale (FP16 or shift-only) can preserve accuracy at substantially lower energy.
- 2) **Two-level scaling is not universally necessary**. For layers with low dynamic range, single-level scaling with an appropriate block size and scale precision can match accuracy while eliminating an entire rescale stage.
- 3) **The optimal configuration is workload-specific**. Small- K , low-kurtosis layers tolerate aggressive precision reduction; large- K , heavy-tailed layers require conservative accumulation. This workload-dependence is the key validation of our framework—it proves no single bundled scheme can be universally optimal.
- 4) We target $\geq 2\times$ **energy reduction** in the rescale/accumulation stages with $<1\%$ accuracy degradation relative to NVFP4 for the workload-optimal configuration.

II. TECHNICAL CONTRIBUTIONS

A. Current Solutions and Related Work

Current 4-bit schemes each bundle a fixed rescale pipeline (Table I).

TABLE I
EXISTING 4-BIT SCHEMES AS FIXED PIPELINE CONFIGURATIONS.

Scheme	# Levels	Scale Fmt	Granularity	Rescale Op
INT4-GPTQ	1	FP16+zp	g=128	FP16 dequant
MXFP4	1	E8M0	b=32	Bit-shift
NVFP4	2	FP8+tensor	b=16+tensor	FP32 $\times 2$
HiFloat4	3	E6M2+E1+E1	hierarchical	Mixed

All recent works share a common pattern: each proposes or optimizes one new fixed design without formulating the problem as a joint search. HiFloat4 fixes its three-level hierarchy as a single format; AXE optimizes accumulator precision in a software-only PTQ framework without hardware cost modeling; MixPE performs Pareto DSE over dequantization *placement* but does not touch the internal rescale pipeline structure; M²XFP [5] and MX+ [6] enhance MX through metadata augmentation, each producing one new fixed scheme. Hardware-aware quantization (HAQ, mixed-precision NAS) searches over per-layer bit-width on *fixed* hardware, not over the MMA pipeline structure.

What’s missing: The problem has never been formulated as a structured DSE—decomposing the rescale pipeline into orthogonal dimensions, defining a tractable search space, and systematically evaluating the joint accuracy–hardware–cost tradeoff.

B. Our Approach: Unbundling the Rescale Pipeline

We decompose the 4-bit MMA datapath into independently tunable dimensions. We fix weight format (E2M1) and activation format (FP8 E4M3) and focus on the three core rescale-pipeline dimensions:

Dimension 1: Rescale pipeline structure—the number of rescale levels (0, 1, or 2) and block granularity at each level ($b \in \{16, 32, 64, \text{per-channel}\}$; for 2-level, coarse $B \in \{256, \text{per-channel, per-tensor}\}$). Finer granularity and more levels improve accuracy but add rescale operations and scale-factor storage.

Dimension 2: Scale-factor format (per level, independently)—FP32, FP16, BF16, FP8 E4M3, E8M0 (power-of-2 bit-shift, nearly free), or fixed-point. In a 2-level pipeline the fine and coarse levels may use different formats, e.g., E8M0 shift at fine level + FP16 at coarse level.

Dimension 3: Accumulator precision—FP32, FP16, BF16, INT32, or segmented (accumulate N partial products in FP16, then reduce to FP32).

After pruning invalid combinations (e.g., accumulator precision $\geq$ operand precision), we estimate ~ 50 viable configurations.

1) *Locating existing schemes:* Every existing scheme is a single hand-designed point in this space:

- **NVFP4** = {2-level (b=16, B=tensor), FP32 rescale $\times 2$, FP32 acc}
- **MXFP4** = {1-level (b=32), E8M0 shift, FP32 acc}
- **INT4-GPTQ** = {1-level (g=128), FP16 dequant, FP32 acc}
- **HiFloat4** = {3-level (hierarchical), E6M2+E1+E1, FP32 acc}

Our contribution is the **formulation itself**: decomposing the bundled pipeline into this search space, building the evaluation infrastructure (PyTorch accuracy emulation + AccelForge hardware cost modeling), and demonstrating that (a) all existing schemes are suboptimal, and (b) the optimal configuration is workload-dependent—validating the necessity of this DSE framework.

C. Tradeoffs and Challenges

- **Search space tractability:** Pruning reduces the combinatorial space to ~ 50 configurations, manageable for exhaustive sweep.
- **Design-time vs. runtime:** The accelerator fixes the pipeline at design time; a single configuration must cover all FFN layers, or we allow 2–3 switchable modes with accounted switching cost.
- **Scale-factor storage overhead:** Finer blocks increase scale-factor storage, reducing effective compression. This overhead is included in total efficiency metrics.

III. EVALUATION

A. Baselines and Fair Comparison

We compare against INT4, MXFP4, NVFP4, and HiFloat4. All configurations use 4-bit weights with FP8 E4M3 activations; effective compression ratio (including scale-factor overhead) is reported. GEMM shapes are identical across configurations. Area is held constant or included via energy-delay-area product (EDAP). Accuracy is measured as per-layer output MSE relative to FP16 and end-to-end perplexity when feasible.

B. Experiment 1: Energy Breakdown of Existing Schemes

Goal: Quantify the energy cost of each MMA pipeline stage for baseline schemes, establishing that the rescale/accumulation overhead is significant and worth optimizing.

Method: Model a systolic-array MMA unit in AccelForge for each baseline. Decompose energy into: operand fetch, multiply, accumulation, rescale (per level), scale-factor fetch, and writeback.

Class concepts: Energy cost hierarchy (compute vs. data movement); multiplier energy scaling with bit-width.

Metrics & visualization: Stacked bar chart—X-axis: scheme, Y-axis: energy per output element (pJ), segmented by pipeline stage. Area per MAC (μm^2) reported alongside.

Expected trend: Our preliminary results (Figure 1) already confirm that NVFP4’s rescale stages dominate at 67% of energy. Extending this analysis to INT4 and MXFP4 will show that MXFP4’s shift-based rescale is nearly free, while INT4 falls in between, establishing the motivation for rescale pipeline optimization.

C. Experiment 2: Accuracy–Efficiency Pareto Frontier

Goal: Sweep the design space and demonstrate that the Pareto frontier contains configurations dominating all existing bundled schemes.

Method: (1) *Accuracy:* PyTorch emulator with configurable rescale pipeline; evaluate on FFN layers from Llama-2-7B and ViT-L; measure per-layer MSE. (2) *Hardware cost:* AccelForge model for each configuration; extract energy per output element and area. (3) Overlay all configurations and baselines on the accuracy–energy plane.

Class concepts: Design-space exploration; Pareto optimality; precision–efficiency tradeoff.

Metrics & visualization: Scatter plot—X-axis: energy/output (pJ), Y-axis: negative MSE. Pareto frontier curve highlighted; existing schemes as labeled reference points.

Expected trend: Configurations such as {1-level b=16, E8M0 fine shift, FP16 acc} or {2-level b=16+channel, E8M0 fine + FP16 coarse, BF16 acc} will match NVFP4 accuracy at lower energy. All four existing schemes will be interior to the Pareto frontier, demonstrating the value of unbundled pipeline design.

D. Experiment 3: Workload-Specificity of Optimal Configuration

Goal: Show that the optimal pipeline configuration varies with workload characteristics (K , weight distribution), validating that the DSE framework is *necessary*—not just that better points exist, but that no single point is universally best.

Method: Select 3 configurations from the Pareto frontier (aggressive / moderate / conservative). Evaluate each on every FFN GEMM layer from Llama-2-7B and ViT-L, varying in K and weight distribution. Map the per-layer optimal configuration to the feature space.

Class concepts: Workload characterization; accumulation chain length vs. required dynamic range; design-time specialization vs. one-size-fits-all.

Metrics & visualization: (a) 2D scatter— $X=K$, Y =kurtosis, color=optimal configuration, revealing decision boundaries. (b) Table: percentage of layers where each configuration wins, by model. (c) Energy savings from per-layer-optimal assignment vs. best single fixed configuration.

Expected trend: Small- K layers tolerate aggressive precision reduction (shift-only rescale, FP16 accumulator); large- K layers require conservative settings. K will be the strongest predictor. Cross-model analysis will confirm that workload features—not model identity—determine the optimal configuration.

IV. TIMELINE AND WORK DISTRIBUTION

Week	Tasks	Owner
1–2	Define/prune search space; PyTorch MMA emulator	Joint
3–4	AccelForge models; Experiment 1 (energy breakdown)	Joint
5–6	Full DSE sweep; Pareto frontier (Experiment 2)	Joint
7–8	Workload sensitivity (Experiment 3); write-up	Joint

Work split: One member focuses on the software accuracy pipeline (PyTorch emulation, model profiling). The other focuses on hardware cost modeling (AccelForge configuration, energy/area estimation). Both collaborate on Pareto analysis, workload sensitivity analysis, and writing.

REFERENCES

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